

SPARSE WAVELET ESTIMATION OF ADDITIVE FUNCTIONAL COEFFICIENTS

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Abstract: We study a varying coefficient model in which the effect of each linear covariate is additive in several modulating covariates, and each additive component is expanded in a periodized wavelet basis. The resulting design is a sieve of products between a linear covariate and a wavelet evaluated at a modulator, and the components are estimated by a single lasso problem in which the levels of the coefficients are left unpenalized. Three results are established. The population Gram matrix of the product design has its smallest eigenvalue bounded away from zero under a conditional second moment condition on the linear covariates and a bounded joint density for the modulators, with no independence between the two groups of covariates; the estimator satisfies a nonasymptotic oracle inequality that needs no cone condition, because the unpenalized levels are profiled out; and, at a resolution level chosen inside an explicit window, the prediction error and the componentwise error attain the rate of nonlinear wavelet approximation up to a logarithmic factor, under the same Besov condition that governs the linear sieve. The last result rests on the observation that the Besov condition already implies compressibility of the sieve coefficients, so that adaptivity costs

no additional assumption. The estimator is a single convex program and is fit by coordinate descent on a sparse design matrix.

Key words and phrases: Additive coefficient model, Besov space, effective dimension, lasso, oracle inequality, varying coefficient model, wavelets.

1. Introduction

Applied problems often present an effect that is not constant across the sample. The return to schooling changes with age and with experience, the effect of temperature on electricity demand changes with the hour of the day and with humidity, and the effect of a treatment changes with the age and the body mass index of the patient. The varying coefficient model of Hastie and Tibshirani (1993) and Cai, Fan, and Yao (2000) accommodates this by letting the coefficient of each covariate be a function of a modulating variable; see Fan and Zhang (2008) for a review. When more than one variable modulates the effect, a fully nonparametric coefficient runs into the curse of dimensionality, and the standard remedy is to make each coefficient additive in the modulators, which is the additive coefficient model of Xue and Yang (2006), studied further by Xue and Liang (2010) and Liu and Yang (2010).

The estimators available for this model are built from polynomial splines

and from kernels, and are governed by a global smoothness index, so that a component with a peak, a jump or a change of regime is either oversmoothed or paid for everywhere in the domain. Wavelet bases combined with an ℓ_1 penalty are the standard answer to this difficulty, because a function whose smoothness is inhomogeneous is still economically represented by a few large coefficients (Donoho and Johnstone, 1994, 1998; Antoniadis and Fan, 2001). They have been used for varying coefficients without an additive structure and without penalization (Zhou and You, 2004; Montoril, Morettin, and Chiann, 2018), and for additive models without a multiplying covariate (Sardy and Tseng, 2004; Zhang and Wong, 2003; Haris et al., 2018; Amato et al., 2022; Sardy and Ma, 2024); penalized estimation in the additive coefficient model itself has been carried out with grouped penalties on spline bases (Wei et al., 2011; Antoniadis et al., 2014; Ma et al., 2015), again under global smoothness.

Klopp and Pensky (2015) study the varying coefficient model in which a single index modulates every coefficient. They expand each coefficient function in an orthonormal basis, estimate the resulting array by a block lasso, and obtain a nonasymptotic oracle inequality, a Besov rate that adapts to inhomogeneous smoothness, and a matching minimax lower bound, under the assumption that the covariates are independent of the index. The

present paper carries that programme to the situation an applied problem usually presents, in which several variables modulate a coefficient at once. We let each coefficient be additive in the modulators, which keeps the estimator free of the curse of dimensionality in their number, we allow the covariates to depend on the modulators, and we penalize coefficient by coefficient rather than in blocks, leaving the levels of the coefficients unpenalized. Each of these costs something. Additivity produces cross terms between distinct modulators, so the population Gram matrix no longer factorizes as a Kronecker product and has to be bounded directly; dependence between the covariates and the modulators replaces a marginal second moment by a conditional one; and the unpenalized levels leave a term in the risk that block penalization does not incur. Our main result, Theorem 1, states that the estimator attains the rate of nonlinear wavelet approximation up to a logarithmic factor, under the same Besov assumption that governs the linear sieve, and it reduces to the rate of Klopp and Pensky when a single modulator is present.

We call the resulting procedure WAFC, for wavelet additive functional coefficients. Its contributions are of three kinds. The first is the theory of the additive product design: a design condition for the Gram matrix of the products between a linear covariate and a wavelet of a modulator,

which carries a cross term between distinct modulators and does not assume independence (Proposition 2); an oracle inequality that needs no cone condition, because the unpenalized levels are profiled out (Theorem 2); and rates obtained by compressibility (Theorem 1). The second is the reading that compressibility costs no assumption: the Besov condition on the components already implies that their wavelet coefficients lie in a weak ℓ_τ ball with $\tau = (s + 1/2)^{-1}$ (Lemma 3), and it is this that separates the rate of the estimator from the rate of the linear sieve when the Besov integrability index is smaller than two. The third is computational: the estimator is one convex program on a sparse design matrix, and the resolution level and the penalty are chosen jointly.

The rest of the paper is organized as follows. Section 2 introduces the model, its identification, the wavelet sieve and the estimator. Section 3 states the assumptions and the theory, opening with the main rate and then giving the three results it rests on. Section 4 describes the computation, the scale of the penalty and the tuning rules. Proofs are collected in the online Supplementary Material.

Throughout, $\|a\|_\pi$ is the ℓ_π norm of a vector, $\|h\|_n^2 = n^{-1} \sum_{i \leq n} h_i^2$ is the empirical norm, $\lambda_{\min}(\cdot)$ and $\lambda_{\max}(\cdot)$ are the extreme eigenvalues of a symmetric matrix, $\|\cdot\|_{\text{op}}$ is the operator norm, and $a_n \asymp b_n$ means that

a_n/b_n is bounded away from zero and infinity. For a support S , δ_S is the vector that agrees with δ on S and vanishes elsewhere.

2. Model and Wavelet Sieve

2.1 The additive functional coefficient model

Let Y be a scalar response, $\mathbf{X} = (X_1, \dots, X_p)^\top$ a vector of linear covariates and $\mathbf{U} = (U_1, \dots, U_q)^\top$ a vector of modulating covariates, taking values in $[0, 1]^q$ after a monotone transformation of each coordinate. The model is

$$Y = \sum_{\ell=1}^p \beta_\ell(\mathbf{U}) X_\ell + \varepsilon, \quad \mathbb{E}(\varepsilon \mid \mathbf{X}, \mathbf{U}) = 0, \quad (2.1)$$

with each functional coefficient additive in the modulators,

$$\beta_\ell(u) = c_\ell + \sum_{m=1}^q g_{\ell m}(u_m), \quad \int_0^1 g_{\ell m}(v) dv = 0, \quad \ell = 1, \dots, p. \quad (2.2)$$

The constant c_ℓ is the level of the ℓ th coefficient and $g_{\ell m}$ is the component of that coefficient along the m th modulator. The centering in (2.2) is with respect to Lebesgue measure, which is the convention of the wavelet literature and the one the basis of Section 2.2 imposes at no cost; the parametrization centered at the law of $\mathbf{U}$ is obtained from it by the shift $c_\ell \mapsto c_\ell + \sum_m \mathbb{E}\{g_{\ell m}(U_m)\}$, which moves mass between the levels and the components but changes neither β_ℓ nor the estimator.

2.1 The additive functional coefficient model

The model contains several familiar ones. With $X_1 \equiv 1$ the first coefficient is an additive intercept, so that $p = 1$ gives the additive model and $p > 1$ with $g_{\ell m} \equiv 0$ for $\ell \geq 2$ gives the partially linear additive model; $q = 1$ gives the classical varying coefficient model with a single index; and $g_{\ell m} \equiv 0$ for all pairs gives linear regression. Writing $f(x, u) = \sum_{\ell} x_{\ell} \beta_{\ell}(u)$ for the regression function, the first question is whether the distribution of $(Y, \mathbf{X}, \mathbf{U})$ determines the parameters. Two degeneracies have to be excluded: if $\mathbf{X}$ is confined to a hyperplane given $\mathbf{U}$, only a combination of the coefficients is determined, and if one modulator is a function of another, only the sum of the corresponding components is.

Proposition 1 (Identification). *Suppose that $E \|\mathbf{X}\|_2^2 < \infty$, that $E(\mathbf{X}\mathbf{X}^{\top} \mid \mathbf{U})$ is nonsingular almost surely, and that $\mathbf{U}$ has a Lebesgue density on $[0, 1]^q$ that is positive almost everywhere. Then, among the parametrizations that satisfy (2.2), the levels $c_1, \dots, c_p$ and the components $g_{\ell m} \in L_2[0, 1]$ are determined by f : the levels uniquely, and each component up to a Lebesgue null set.*

Dependence between $\mathbf{X}$ and $\mathbf{U}$ is allowed, and so is dependence among the modulators; what the two conditions exclude is a linear combination of the linear covariates that is a function of the modulators, and a modulator that is a function of the others. Both exclusions return in quantitative form

in Section 3.

2.2 The wavelet sieve

Let ϕ and ψ be the scaling function and the mother wavelet of an orthonormal multiresolution analysis of $L_2(\mathbb{R})$, compactly supported, with ψ bounded and with N vanishing moments, and let L be the length of the filter, so that the support of ψ is contained in $[0, L - 1]$. Write

$$\psi_{jk}(u) = \sum_{\iota \in \mathbb{Z}} 2^{j/2} \psi\{2^j(u + \iota) - k\}, \quad j \geq 0, \quad k = 0, \dots, 2^j - 1, \quad (2.3)$$

for the periodization of the basis to $[0, 1]$. The system $\{\mathbf{1}\} \cup \{\psi_{jk}\}$ is an orthonormal basis of $L_2[0, 1]$ under Lebesgue measure and every wavelet in it integrates to zero (Daubechies, 1992, Section 9.3). We take the coarsest level to be $j_0 = 0$, so that the only scaling function is the constant $\phi_{00} \equiv 1$, and we discard it: it is already carried by the level c_ℓ . The sieve of resolution J is then

$$W_J = \text{span}\{\psi_{jk} : 0 \leq j < J, 0 \leq k < 2^j\}, \quad \dim W_J = N_J = 2^J - 1,$$

the orthogonal complement of the constants inside the multiresolution space V_J , and Π_J is the orthogonal projection onto it. Every element of W_J integrates to zero, so that the identification constraint of (2.2) holds by construction, no numerical restriction is imposed on the fit, and the sieve

is a plain linear space.

Lemma 1 (The basis carries the constraint). *Under the conditions of Proposition 1, for every J the map*

$$(\mathbf{c}, \boldsymbol{\theta}) \mapsto \sum_{\ell=1}^p x_{\ell} \left\{ c_{\ell} + \sum_{m=1}^q \sum_{j < J} \sum_k \theta_{\ell m, jk} \psi_{jk}(u_m) \right\}$$

is injective on $\mathbb{R}^p \times \mathbb{R}^d$, $d = pqN_J$; equivalently, the population Gram matrix of the design defined in (2.4) is positive definite. If instead the column $X_{\ell}\phi_{00}(U_m)$ is retained in each block, that matrix is singular with nullity exactly pq .

Replacing $g_{\ell m}$ by $\Pi_J g_{\ell m}$ leaves a bias, which Section 3 controls under a Besov condition. Two practical consequences of periodization deserve mention. A component whose values at the two ends of the interval disagree is not periodic, and its projection error then stalls at the order $2^{-J/2}$ in L_2 regardless of the interior smoothness (Proposition S3.1 in the Supplementary Material). The two remedies, rescaling the modulators to a proper subinterval and using the basis of Cohen, Daubechies, and Vial (1993) adapted to the interval, are matters of computation and are discussed in Section 4.

2.3 The estimator

Let $(Y_i, \mathbf{X}_i, \mathbf{U}_i)$, $i = 1, \dots, n$, be independent copies of $(Y, \mathbf{X}, \mathbf{U})$. The row of the design matrix is the vector of the p linear covariates followed by the

$d = pqN_J$ products between a linear covariate and a wavelet of a modulator,

$$\mathbf{Z} = [\mathbf{A} \ \mathbf{B}], \quad \mathbf{A}_{i\ell} = X_{i\ell}, \quad \mathbf{B}_{i,(\ell m, jk)} = X_{i\ell} \psi_{jk}(U_{im}), \quad (2.4)$$

with the blocks (ℓ, m) in lexicographic order and, inside a block, the wavelets ordered by level and then by translation. Writing $\mathbf{Y} = (Y_1, \dots, Y_n)^\top$, the estimator is

$$(\hat{\mathbf{c}}, \hat{\boldsymbol{\theta}}) \in \arg \min_{(\mathbf{c}, \boldsymbol{\theta}) \in \mathbb{R}^p \times \mathbb{R}^d} \left\{ \|\mathbf{Y} - \mathbf{A}\mathbf{c} - \mathbf{B}\boldsymbol{\theta}\|_n^2 + 2\lambda \|\boldsymbol{\theta}\|_1 \right\}. \quad (2.5)$$

The penalty does not reach the levels $\mathbf{c}$: they are p parameters of a parametric part, and shrinking them would bias every fitted coefficient by a constant. The fitted components and coefficients are recovered from $\hat{\boldsymbol{\theta}}$ by

$$\hat{g}_{\ell m}(v) = \sum_{j < J} \sum_k \hat{\theta}_{\ell m, jk} \psi_{jk}(v), \quad \hat{\beta}_\ell(u) = \hat{c}_\ell + \sum_{m=1}^q \hat{g}_{\ell m}(u_m).$$

Problem (2.5) is a lasso with some coordinates unpenalized, so it is convex, and it is strictly convex, hence uniquely solved, whenever the Gram matrix of $\mathbf{Z}$ is nonsingular. Two structural variants are available at no conceptual cost and are discussed in Section 4: a sparse group lasso (Simon et al., 2013) whose groups are the blocks (ℓ, m) , which removes an entire component when a modulator does not act on a coefficient, and the block lasso of Klopp and Pensky (2015). The theory below is for (2.5).

3. Theory

3.1 Assumptions

Assumption 1 (Sampling and error). The observations $(Y_i, \mathbf{X}_i, \mathbf{U}_i)$, $i \leq n$, are independent and identically distributed and follow (2.1), and the error is conditionally sub-Gaussian: there is $\sigma < \infty$ with $\mathbb{E}\{\exp(t\varepsilon_i) \mid \mathbf{X}_i, \mathbf{U}_i\} \leq \exp(\sigma^2 t^2/2)$ almost surely, for all real t .

Assumption 2 (Linear covariates). $\max_\ell |X_\ell| \leq B_X$ almost surely, and there are constants $0 < \kappa_1 \leq \kappa_2 < \infty$ with $\kappa_1 \leq \lambda_{\min}\{\mathbb{E}(\mathbf{X}\mathbf{X}^\top \mid \mathbf{U})\} \leq \lambda_{\max}\{\mathbb{E}(\mathbf{X}\mathbf{X}^\top \mid \mathbf{U})\} \leq \kappa_2$ almost surely.

Assumption 3 (Modulating covariates). $\mathbf{U}$ takes values in $[0, 1]^q$ and has a joint Lebesgue density $p_{\mathbf{U}}$ with $0 < c_U \leq p_{\mathbf{U}}(u) \leq C_U < \infty$ for every $u \in [0, 1]^q$.

Assumption 4 (Basis). The wavelets are the periodization (2.3) of an orthonormal, compactly supported family with ψ bounded and N vanishing moments, and $j_0 = 0$.

Assumption 5 (Besov regularity). There are $s > 0$, $1 \leq \pi, r \leq \infty$ and $C_g < \infty$ such that every component satisfies $\int_0^1 g_{\ell m} = 0$ and, writing $\theta_{\ell m, jk}^* = \langle g_{\ell m}, \psi_{jk} \rangle$ and $\theta_{\ell m, j}^*$ for the vector of the coefficients of level j ,

$$\|\theta_{\ell m, j}^*\|_\pi \leq C_g 2^{-j(s+1/2-1/\pi)} \quad \text{for every } j \geq 0.$$

The effective regularity $s' = s - (1/\pi - 1/2)_+$ is positive.

Assumptions 2 and 3 are the quantitative forms of the two conditions of Proposition 1. In the second it is the joint density that is bounded below, not the marginals: if $U_2 = U_1$ almost surely both marginals are uniform and the design is singular. Assumption 5 is taken as a condition on the sequence of coefficients rather than as membership in a function space, as in Klopp and Pensky (2015); when ψ has more than s vanishing moments and enough Hölder regularity the two are equivalent, on the circle (Cohen, 2003, Section 3.6), and the implication that the proofs use needs only the vanishing moments. The index s' is the regularity that a linear sieve sees and s is the one that nonlinear approximation sees; they differ exactly when $\pi < 2$, which is the regime of inhomogeneous smoothness.

3.2 The main result

Write $\widehat{f}_i = \mathbf{A}_i \widehat{\mathbf{c}} + \mathbf{B}_i \widehat{\boldsymbol{\theta}}$ for the fit, d_n for the number of penalized columns at resolution J_n , and fix a confidence level $\alpha \in (0, 1)$. The penalty is the one that the oracle inequality of Section 3.4 calls for, namely

$$\lambda_n = 2\sigma B_X \sqrt{2C_U} \sqrt{\frac{2 \log(2d_n/\alpha)}{n}} \asymp \sigma B_X \sqrt{C_U} \sqrt{\frac{J_n}{n}}. \quad (3.1)$$

Theorem 1 (Rate of nonlinear approximation). *Let Assumptions 1 to 5 hold with p, q and all the constants fixed as n grows, and set $\tau = (s+1/2)^{-1}$.*

Suppose $s' > s/(2s + 1)$ and take $J_n = \lceil c \log_2 n \rceil$ with

$$\max\left\{\frac{\tau}{2}, \frac{2-\tau}{4s'}\right\} \leq c < 1, \quad (3.2)$$

a window that is nonempty precisely under that condition. Then, with λ_n

as in (3.1),

$$\left\|\widehat{f} - f\right\|_n^2 = O_p\left\{\left(\frac{\log n}{n}\right)^{2s/(2s+1)}\right\}, \quad \sum_{\ell,m} \left\|\widehat{g}_{\ell m} - g_{\ell m}\right\|_{L_2[0,1]}^2 = O_p\left\{\left(\frac{\log n}{n}\right)^{2s/(2s+1)}\right\}.$$

Three readings of Theorem 1 matter. First, the exponent is the one of nonlinear wavelet approximation, $2s/(2s + 1)$, and not the one of the linear sieve, $2s'/(2s' + 1)$ of Theorem 3 below; the two agree when $\pi \geq 2$ and separate when $\pi < 2$, which is where the ℓ_1 penalty pays for itself. Second, the rate is attained under Assumption 5 alone: no compressibility is assumed on top of it, because the Besov condition already implies it, with $\tau = (s + 1/2)^{-1}$, by Lemma 3. Third, the rate $(\log n/n)^{2s/(2s+1)}$ is the minimax reference of the Gaussian sequence model over weak ℓ_τ balls with per coordinate noise of order λ_n (Donoho and Johnstone, 1998); the logarithmic factor is the price of the single, nonadaptive penalty level (3.1), which takes a union bound over the d_n columns. No lower bound is proved here for $q \geq 2$: the comparison is with the reference of the sequence model, and the minimax lower bound of Klopp and Pensky (2015) covers the case of a single modulator with covariates independent of it.

3.3 Approximation and the design condition

The proof combines three ingredients, given below: the approximation error of the sieve, the design condition, and a nonasymptotic oracle inequality. Section 3.5 assembles them, first at a dense reading of the estimation term, which gives the sieve rate, and then at a sparse comparator, which gives Theorem 1.

3.3 Approximation and the design condition

Lemma 2 (Bias of the sieve). *Under Assumptions 4 and 5, for every pair (ℓ, m) and every $J \geq 0$,*

$$\|g_{\ell m} - \Pi_J g_{\ell m}\|_{L_2[0,1]}^2 = \sum_{j \geq J} \|\theta_{\ell m, j}^*\|_2^2 \leq \frac{C_g^2}{1 - 2^{-2s'}} 2^{-2Js'}.$$

If in addition Assumptions 2 and 3 hold, then, with $f_J(x, u) = \sum_{\ell} x_{\ell} \{c_{\ell} + \sum_m (\Pi_J g_{\ell m})(u_m)\}$,

$$\|f - f_J\|_{L_2(P)}^2 \leq \frac{(pq)^2 B_X^2 C_U C_g^2}{1 - 2^{-2s'}} 2^{-2Js'} =: \mathcal{B}_J^2. \quad (3.3)$$

The exponent is s' and not s , and this is where the loss $(1/\pi - 1/2)_+$ enters: inside a level, an ℓ_{π} bound with $\pi < 2$ controls the ℓ_2 norm without gain. The bias uses only the bound on the linear covariates and the marginal density of each modulator, so dependence between $\mathbf{X}$ and $\mathbf{U}$ does not affect it. What it does affect is the passage from the prediction norm to the norm of each component, which is the design condition.

3.3 Approximation and the design condition

Let $\Psi(u) \in \mathbb{R}^K$, $K = 1 + qN_J$, collect the constant and the wavelets of the q modulators, so that the row of $\mathbf{Z}$ in (2.4) is a permutation of the Kronecker product $\mathbf{X} \otimes \Psi(\mathbf{U})$. Write $\Sigma = \mathbb{E}(\mathbf{Z}\mathbf{Z}^\top)$ and $\hat{\Sigma} = \mathbf{Z}^\top \mathbf{Z}/n$ for the population and empirical Gram matrices and $\Sigma_\Psi = \mathbb{E}\{\Psi(\mathbf{U})\Psi(\mathbf{U})^\top\}$ for the part in the modulators alone.

Proposition 2 (Gram matrix of the product design). *Let Assumptions 2 to 4 hold and write $D = p + d$ and $\gamma = \kappa_1 c_U$.*

(i) $c_U \leq \lambda_{\min}(\Sigma_\Psi) \leq \lambda_{\max}(\Sigma_\Psi) \leq C_U$ for every J , and consequently $\gamma \leq \lambda_{\min}(\Sigma) \leq \lambda_{\max}(\Sigma) \leq \kappa_2 C_U$. If $\mathbf{X}$ and $\mathbf{U}$ are independent, Σ is, up to the permutation of columns, the Kronecker product $\mathbb{E}(\mathbf{X}\mathbf{X}^\top) \otimes \Sigma_\Psi$, and the two bounds are attained as products of eigenvalues.

(ii) With $R_J = pB_X^2(1 + qC_\psi 2^J)$, where C_ψ depends only on the wavelet family, $\|\mathbf{Z}_i\|_2^2 \leq R_J$ almost surely and

$$\mathbb{P}\left(\lambda_{\min}(\hat{\Sigma}) \geq \frac{\gamma}{2}\right) \geq 1 - 2D \exp\left\{-\frac{n\gamma^2}{8R_J(\kappa_2 C_U + \gamma/3)}\right\},$$

which tends to one whenever $pq2^J \log(pq2^J)/n \rightarrow 0$.

Part (i) is stated for the general case: independence between the covariates and the modulators buys the exact factorization but not a better constant. A lower bound on the smallest eigenvalue dominates every compatibility and restricted eigenvalue constant, on every support and with no

3.4 A nonasymptotic oracle inequality

cone restriction, since $\delta^\top A \delta \geq \lambda_{\min}(A) \|\delta\|_2^2 \geq \lambda_{\min}(A) \|\delta_S\|_1^2 / |S|$, and it is in this form that the next subsection consumes it. In the resolution regime of Section 3.5 the requirement of part (ii) is of the same order as the rate itself, so it costs nothing there; if pq were larger than $n^{1/2}$, only the compatibility constant could be transferred, through the entrywise deviation (Bühlmann and van de Geer, 2011, Corollary 6.8), at the price of a factor s_0^2 and of the cone.

3.4 A nonasymptotic oracle inequality

Let $\boldsymbol{\beta}^* = (\mathbf{c}, \boldsymbol{\theta}^*)$ be the sieve oracle, that is the vector whose components are the levels and the coefficients $\theta_{\ell m, jk}^*$ of Assumption 5 for $j < J$, so that f_J is its fit, and let

$$\mathbf{b} = f - f_J \in \mathbb{R}^n, \quad S_0 = \text{supp}(\boldsymbol{\theta}^*), \quad s_0 = |S_0|, \quad v = \widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}^*.$$

Let $\Pi_{\mathbf{A}}$ be the orthogonal projection onto the column space of $\mathbf{A}$, which exists whenever $\mathbf{A}$ has rank p , let $M_{\mathbf{A}} = I_n - \Pi_{\mathbf{A}}$, and let $\tilde{\mathbf{B}} = M_{\mathbf{A}} \mathbf{B}$ and $\tilde{\gamma} = \lambda_{\min}(\tilde{\mathbf{B}}^\top \tilde{\mathbf{B}}/n)$ be the residualized design and its smallest eigenvalue. Finally let $\hat{\sigma}_{\max}^2 = \max_a (\hat{\Sigma})_{aa}$ be the largest empirical norm of a penalized column.

Theorem 2 (Oracle inequality). *Let Assumption 1 hold, suppose $\mathbf{A}$ has*

3.4 A nonasymptotic oracle inequality

rank p , and take $\lambda \geq 2\lambda_0$ with $\lambda_0 = \sigma\widehat{\sigma}_{\max}\{2\log(2d/\alpha)/n\}^{1/2}$. On the event

$\mathcal{T} = \{\|\widetilde{\mathbf{B}}^\top \boldsymbol{\varepsilon}/n\|_\infty \leq \lambda/2\}$, which has probability at least $1 - \alpha$:

(i) with no condition on the design, $\|\widetilde{\mathbf{B}}v\|_n^2 \leq 6\lambda \|\boldsymbol{\theta}^*\|_1 + 4 \|\mathbf{b}\|_n^2$;

(ii) if $\widetilde{\gamma} > 0$,

$$\|\widetilde{\mathbf{B}}v\|_n^2 \leq \frac{64\lambda^2 s_0}{\widetilde{\gamma}} + 16 \|\mathbf{b}\|_n^2, \quad \|v\|_2^2 \leq \frac{64\lambda^2 s_0}{\widetilde{\gamma}^2} + \frac{16 \|\mathbf{b}\|_n^2}{\widetilde{\gamma}};$$

(iii) in either regime, $\|\widehat{f} - f\|_n \leq \|\widetilde{\mathbf{B}}v\|_n + 2 \|\mathbf{b}\|_n + \|\Pi_{\mathbf{A}}\boldsymbol{\varepsilon}\|_n$, and $\mathbb{E}(\|\Pi_{\mathbf{A}}\boldsymbol{\varepsilon}\|_n^2 \mid$

$$\mathbf{X}, \mathbf{U}) \leq \sigma^2 p/n.$$

Three features separate Theorem 2 from the standard treatment (Bühlmann and van de Geer, 2011, Chapter 6). The unpenalized levels are not placed inside the support, which would require a compatibility condition on a cone; they are profiled out, and the Schur complement transfers the full smallest eigenvalue of $\widehat{\Sigma}$ to the residualized block, $\widetilde{\gamma} \geq \lambda_{\min}(\widehat{\Sigma})$, so that Proposition 2(ii) is used as it stands and the constant does not depend on S_0 . The model is misspecified inside the sieve, and the deterministic residual $\mathbf{b}$ is handled by Young's inequality rather than by duality, which keeps the statement nonasymptotic and valid also when the bias dominates. The penalty is calibrated by the empirical column norms and not by $\max_{i,a} |Z_{ia}|$: the former is bounded in probability by $2B_X^2 C_U$, while the latter is of order

$2^{J/2}$ and would inflate λ by that factor and destroy the rates of Section 3.5.

The term $\sigma^2 p/n$ in part (iii) is the price of leaving the p levels out of the penalty; no choice of λ removes it, it is of smaller order than the estimation term whenever $p \lesssim s_0 \log(pqN_J)$, and it is absent from Klopp and Pensky (2015), where the levels are penalized along with everything else.

3.5 Rates

Under Assumption 5 the sieve oracle is generically dense: the coefficients decay but do not vanish, so that $s_0 = d$ and part (ii) of Theorem 2, read at the oracle, delivers the rate of the linear sieve.

Theorem 3 (Rate of the linear sieve). *Let Assumptions 1 to 5 hold with p, q and the constants fixed, and take*

$$J_n = \min\{J \in \mathbb{N} : 2^J \geq (n/\log n)^{1/(2s'+1)}\}, \quad (3.4)$$

so that $J_n = \log_2 n/(2s'+1) + O(\log \log n)$. Then the requirement of Proposition 2(ii) holds along J_n , being itself of the exact order $(\log n/n)^{2s'/(2s'+1)}$, and, with λ_n as in (3.1),

$$\left\| \widehat{f} - f \right\|_n^2 = O_p \left\{ \left(\frac{\log n}{n} \right)^{2s'/(2s'+1)} \right\}.$$

At this resolution the estimation term and the bias term are of the same order, and the term $\sigma^2 p/n$ of Theorem 2(iii) is of strictly smaller order.

Corollary 1 (Components and coefficients). *In the setting of Theorem 3, and with r_n denoting the rate of that theorem,*

$$\sum_{\ell,m} \|\widehat{g}_{\ell m} - g_{\ell m}\|_{L_2[0,1]}^2 = O_p(r_n), \quad \|\widehat{\beta}_\ell - \beta_\ell\|_{L_2([0,1]^q)} = O_p(r_n^{1/2}), \quad \ell = 1, \dots, p,$$

and the same statements hold with the rate of Theorem 1 in place of r_n under the conditions of that theorem.

The passage from the prediction norm to the norm of a single component is where the design condition is used a second time, and the smallest eigenvalue appears squared in the constant. Two further statements complete the picture. The first dispenses with the design condition altogether, at the price of a slower rate, and is the unconditional statement of this paper.

Proposition 3 (Slow rate, no design condition). *Let Assumptions 1, 4 and 5 hold, with $\mathbf{A}$ of rank p and λ_n as in (3.1). Then $\|\widehat{f} - f\|_n^2 = O_p\{\lambda_n \|\boldsymbol{\theta}^*\|_1 + \mathcal{B}_{J_n}^2 + p/n\}$, where $\|\boldsymbol{\theta}^*\|_1 \leq pqC_g \sum_{j < J_n} 2^{j(1/2-s)}$ does not depend on π . In particular, if $s > 1/2$ then $\|\widehat{f} - f\|_n^2 = O_p\{(J_n/n)^{1/2}\}$ for every $J_n \geq \log_2 n/(4s')$; and if $s < 1/2$, the choice $2^{J_n} \asymp (n/J_n)^{1/(1-2s+4s')}$ gives $\|\widehat{f} - f\|_n^2 = O_p\{(J_n/n)^{2s'/(1-2s+4s')}\}$, an exponent that equals $2s/(2s+1)$ when $\pi \geq 2$.*

The second is the approximation lemma that produces Theorem 1. For

a vector $\boldsymbol{\theta}$ let $\theta_{(1)} \geq \theta_{(2)} \geq \dots$ be the nonincreasing rearrangement of the absolute values of its entries and let $\boldsymbol{\theta}^{(M)}$ retain the M largest of them.

Lemma 3 (Besov regularity implies compressibility). *Under Assumptions 4 and 5 with $\pi(s + 1/2) > 1$, the sieve oracle satisfies, for every J and every $k \geq 1$,*

$$\theta_{(k)}^* \leq C_\tau k^{-1/\tau}, \quad \tau = \frac{1}{s + 1/2}, \quad C_\tau = C_g(pq A_{s,\pi})^{s+1/2}, \quad A_{s,\pi} = 1 + \frac{1}{1 - 2^{1-\pi(s+1/2)}},$$

with constants that do not depend on J and with $\tau \in (0, 2)$ for every $s > 0$.

Consequently $\sum_{k>M} \theta_{(k)}^{*2} \leq \tau(2 - \tau)^{-1} C_\tau^2 M^{1-2/\tau}$ for every $M \geq 1$.

Theorem 1 follows by reading Theorem 2(ii) not at the oracle but at the best M term approximation of it, which is legitimate because the proof of that theorem uses the comparator only through the decomposition of $\mathbf{Y}$ and the size of its support. Balancing the two terms that result, $\lambda^2 M / \tilde{\gamma}$ from estimation and $M^{1-2/\tau}$ from the truncation, gives the effective dimension

$$M_n^* \asymp \left(\frac{n}{J_n} \right)^{\tau/2} \ll d_n \asymp pq 2^{J_n}, \quad (3.5)$$

and a bound of order $\lambda_n^{2-\tau}$, which is the rate of Theorem 1. The three inequalities that define the window (3.2) are, in order, that the minimum in M be interior, that the bias of the sieve not dominate, and that the requirement of Proposition 2(ii) hold. Nothing in this argument says that the

estimator selects $M_n^\star$ coordinates: no minimal signal or irrepresentability condition is used, and what compressibility buys is a rate governed by $M_n^\star$ rather than by d_n , not the recovery of a support.

4. Computation and Tuning

4.1 The design matrix and the solver

The design (2.4) is built one block at a time: the $n \times N_J$ matrix of the wavelets of U_m is evaluated once per modulator, by the cascade algorithm of Daubechies and Lagarias (1991), and then multiplied rowwise by X_ℓ . Since a compactly supported wavelet of level j is nonzero on an interval of length $(L-1)2^{-j}$, at most $L-1$ of the 2^j columns of a level are nonzero at a given observation, so a block has at most $n(L-1)J$ nonzero entries out of nN_J and the design is stored in sparse form as soon as J is moderate. Evaluating the basis once on a grid and looking the values up removes the cascade from the inner loop of a simulation.

Problem (2.5) is then a lasso with p unpenalized coordinates, which coordinate descent solves along a path of penalty levels; we use the `glmnet` package. Two points of scale deserve care, because they are what ties the software to the theory. First, the penalty level reported for a fit is the λ of (2.5) and not the one of the solver, which rescales its own by the ratio

between the total number of columns and the number of penalized ones when some coordinates carry zero penalty; the conversion is checked by verifying the Karush-Kuhn-Tucker conditions of (2.5) at every point of the path. Second, when one linear covariate is the constant, its column has zero variance, and the solver may either drop it or share the level with its own intercept; the fit is the same either way, but the level c_ℓ of the constant covariate is only recovered after the intercept is added back to it.

4.2 Rescaling and the boundary

The theory places $\mathbf{U}$ in $[0, 1]^q$ by assumption. In practice each modulator is mapped to that interval by a monotone transformation, and, as in the wavelet estimators of Montoril, Chang, and Vidakovic (2019), the image is $[\epsilon, 1 - \epsilon]$ with ϵ of the order 2^{-J} , which keeps the observations away from the strip where the periodization of (2.3) leaves an artifact; the price is that the fitted component integrates to zero over the rescaled range and not over $[0, 1]$, a shift of level and not of shape. The alternative is the basis adapted to the interval of Cohen, Daubechies, and Vial (1993), which reproduces polynomials on $[0, 1]$ and characterizes Besov regularity without any periodicity (Cohen, 2003, Section 3.9), so that Lemma 2 holds with the Besov space of the interval and the same proof. Its cost is structural:

4.3 Choosing the resolution and the penalty

the coarsest level must be large enough for the two boundary zones to be disjoint, the constant is no longer a single column but a combination of the scaling functions, and the constraint of (2.2) has to be imposed by reparametrization, which leaves $pq(2^{j_0} - 1)$ unpenalized parameters instead of none. We therefore keep the periodized basis as the default and offer the interval basis as an option.

4.3 Choosing the resolution and the penalty

The pair (J, λ) is chosen jointly. The default is K fold cross validation on a grid of resolutions, with the folds drawn once and reused for every J , one design and one penalty path per resolution, and the usual two readings of the curve: the minimizer of the cross validated error and the largest penalty within one standard error of it. Two cheaper rules are also available, the Bayesian information criterion and its extended version, with degrees of freedom equal to the number of nonzero coefficients plus the p levels, which is the usual unbiased count for the lasso.

The theory supplies a fourth rule, since (3.4) and (3.1) are explicit. It is not a practical competitor as it stands, because it needs σ , which has to be estimated, and s' , which is unknown, and because it is a statement about order, so that at moderate sample sizes it can sit above the resolution that

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minimizes the realized error, the pre-asymptotic regime being wide when s' is small. Its interest is as a reference for what the cross validated choice ought to approach, and the difference between the two is a finite sample quantity worth measuring. The four rules are compared in the simulation study.

A remark on the structural variant. Replacing $\|\boldsymbol{\theta}\|_1$ in (2.5) by the sparse group lasso penalty of Simon et al. (2013), with one group per block (ℓ, m) , adds a second tuning parameter and removes entire components rather than isolated coefficients, which is what a reader interested in knowing which modulators act on which coefficients would want. It is a different estimator, with a different oracle inequality (Lounici et al., 2011), and the theory above does not cover it; the choice between the two is settled empirically.

Supplementary Materials

The online Supplementary Material contains the proofs of all the results stated in the paper, together with the auxiliary lemmas they use: the identification argument and the properties of the periodized basis, the approximation lemmas in L_2 and in the supremum norm and the proposition on the cost of periodization, the population and empirical bounds for the Gram

matrix of the product design, the four lemmas behind the oracle inequality, and the two lemmas behind the rates.

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